WORKED SOLUTION

# STEP III 2019, Question 10

Mechanics • Oblique Collisions

**STEP III 2019, Question 10**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

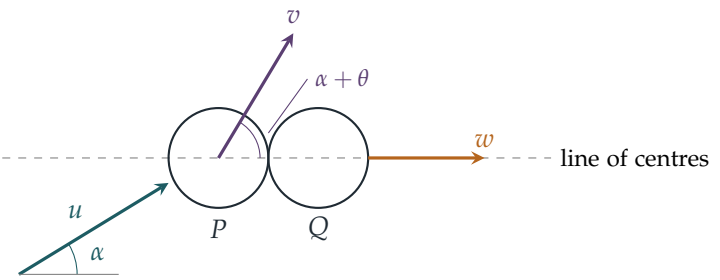
www.ocr.org.uk/Images/696590-2019-past-papers.zip

## The key idea

Split every velocity into a component along the line of centres and a component perpendicular to it. Because the spheres are smooth, the impulse acts only along the line of centres, so $P$'s perpendicular component is unchanged and $Q$ moves off along that line. Conservation of momentum and Newton's experimental law then apply along the line of centres just as in a one-dimensional collision. The rest is trigonometry.

## Solution

Model $P$ and $Q$ as smooth spheres of equal mass $m$ on a smooth horizontal table. Take the line of centres at impact as the $x$-direction, pointing from $P$ to $Q$, and let $y$ be perpendicular to it in the plane of motion, on the side of $P$'s initial velocity.




**Part (i)**

Before the collision $P$ has components $u\cos\alpha$ along the line of centres and $u\sin\alpha$ perpendicular to it.

The contact force between smooth spheres acts along the line of centres. So there is no impulse on $P$ perpendicular to that line, and $P$'s perpendicular component is unchanged. Also, $Q$ starts at rest and receives an impulse only along the line of centres,

so it moves off along that line with speed $w$.

**Ask yourself** Why does $P$'s new direction make the angle $\alpha + \theta$ with the line of centres, and not $\alpha - \theta$?

The impulse on $P$ points backwards along the line of centres. It reduces $P$'s $x$-component, from $u\cos\alpha$ to the value $v_x$ found below, and leaves the $y$-component alone. The velocity therefore swings away from the line of centres, and the angle it makes with that line grows from $\alpha$ to $\alpha + \theta$. We'll confirm below that $v_x > 0$ when $e < 1$, so $0 < \alpha + \theta < \frac{\pi}{2}$.

After the collision, $P$'s components are $v\cos(\alpha+\theta)$ along the line of centres and $v\sin(\alpha+\theta)$ perpendicular to it. Equating the perpendicular components gives

$$u\sin\alpha = v\sin(\alpha+\theta),$$

as required.

Along the line of centres, conservation of momentum (the masses are equal, so we cancel $m$) gives

$$u\cos\alpha = v\cos(\alpha+\theta) + w. \tag{1}$$

Substituting $u = v\sin(\alpha+\theta)/\sin\alpha$ from the first result,

$$\begin{aligned} w &= \frac{v\sin(\alpha+\theta)\cos\alpha}{\sin\alpha} - v\cos(\alpha+\theta) \\ &= \frac{v\big[\sin(\alpha+\theta)\cos\alpha - \cos(\alpha+\theta)\sin\alpha\big]}{\sin\alpha} = \frac{v\sin\theta}{\sin\alpha}, \end{aligned}$$

using the expansion of $\sin\big((\alpha+\theta)-\alpha\big)$. So

$$w = \frac{v\sin\theta}{\sin\alpha}.$$

This has a neat interpretation. Vector momentum conservation says $\mathbf{u} = \mathbf{v} + \mathbf{w}$, and resolving perpendicular to $\mathbf{u}$ gives $0 = v\sin\theta - w\sin\alpha$.

#### Part (ii)

Newton's experimental law along the line of centres says that the speed of separation is $e$ times the speed of approach:

$$w - v\cos(\alpha+\theta) = e\,u\cos\alpha. \tag{2}$$

Substitute $w = v\sin\theta/\sin\alpha$ and $u = v\sin(\alpha+\theta)/\sin\alpha$ from part (i):

$$\frac{v\sin\theta}{\sin\alpha} - v\cos(\alpha+\theta) = \frac{e\,v\sin(\alpha+\theta)\cos\alpha}{\sin\alpha}.$$

Multiplying through by $\sin\alpha/v$ (allowed since $v > 0$ and $\sin\alpha \neq 0$) and rearranging gives

$$\sin\theta = \cos(\theta+\alpha)\sin\alpha + e\sin(\theta+\alpha)\cos\alpha,$$

as required.

To find $\tan\theta$, expand both compound angles:

$$\sin\theta = (\cos\theta\cos\alpha - \sin\theta\sin\alpha)\sin\alpha + e(\sin\theta\cos\alpha + \cos\theta\sin\alpha)\cos\alpha.$$

Collecting the $\sin\theta$ terms on the left:

$$\sin\theta\left(1 + \sin^2\alpha - e\cos^2\alpha\right) = (1+e)\cos\theta\sin\alpha\cos\alpha.$$

Divide by $\cos\theta\cos^2\alpha$ and write $t = \tan\alpha$, using $1/\cos^2\alpha = 1 + t^2$:

$$\tan\theta\left(1 + t^2 + t^2 - e\right) = (1+e)\,t, \qquad \text{so} \qquad \tan\theta = \frac{(1+e)\tan\alpha}{1 - e + 2\tan^2\alpha}.$$

The denominator is positive because $e < 1$, so $\tan\theta > 0$ and $\theta$ really is an acute deflection.

#### The maximum of $\tan\theta$

As $\alpha$ ranges over acute angles, $t = \tan\alpha$ takes every value in $(0, \infty)$. For $t > 0$ we can divide top and bottom by $t$:

$$\tan\theta = \frac{1+e}{\dfrac{1-e}{t} + 2t}.$$

**?** **Ask yourself** How do we maximise this without calculus?

Make the denominator as small as possible. Both $(1-e)/t$ and $2t$ are positive, so by the AM–GM inequality

$$\frac{1-e}{t} + 2t \geq 2\sqrt{\frac{1-e}{t}\cdot 2t} = 2\sqrt{2(1-e)},$$

with equality exactly when $(1-e)/t = 2t$, that is, when $\tan\alpha = \sqrt{(1-e)/2}$. This is a genuine acute angle, so the minimum is attained. Therefore the maximum value of $\tan\theta$ is

$$\tan\theta_{\max} = \frac{1+e}{2\sqrt{2(1-e)}}.$$

(Differentiating with respect to $t$ gives the same stationary point, $2t^2 = 1 - e$.)

## Checks

- Elastic-like limit $e \to 1$: the spheres exchange their components along the line of centres, so $P$ keeps only $u\sin\alpha$ and moves at right angles to the line of centres. Then $\alpha + \theta = \frac{\pi}{2}$, and $\tan\theta = \cot\alpha$. The formula gives $\tan\theta = 2t/(2t^2) = 1/t$, which agrees. The maximum blows up, as it should, because $\theta \to \frac{\pi}{2}$ when $\alpha \to 0$.

- With $e = \frac{1}{2}$ the maximum is $\frac{3/2}{2\sqrt{1}} = \frac{3}{4}$, attained at $\tan\alpha = \frac{1}{2}$ (about $26.6°$). Scanning $\alpha$ finely confirms 0.75. A computer simulation of many random collisions also reproduced all four results of the question.
- Small $\alpha$ (nearly head-on): $\tan\theta \approx \frac{1+e}{1-e}\alpha \to 0$, so a head-on collision causes no deflection, as expected.

## Common slips

- Applying Newton's experimental law to whole speeds instead of to the components along the line of centres. Restitution acts only in the direction of the impulse.
- Writing $P$'s new angle as $\alpha - \theta$. The impulse removes some of the component along the line of centres, so $P$ turns away from that line.
- Forgetting that $e < 1$ is what makes $1 - e + 2\tan^2\alpha$ positive, and that $\tan\alpha$ ranges over all of $(0, \infty)$, so the AM–GM equality case is actually achieved.

## Beyond STEP

Oblique collisions of smooth spheres are the simplest example of scattering. In this problem the same two conservation laws are applied along and across a line, and that idea scales up to particle physics and the kinetic theory of gases. For elastic collisions of equal masses the result is the famous right angle between the outgoing velocities. In the centre-of-mass frame every collision just rotates (and, here, shrinks) the relative velocity, which is a slicker way to see all of these results. Kleppner and Kolenkow's *An Introduction to Mechanics* develops collisions from this viewpoint.

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